

AI 3000: Reinforcement Learning

Tutorial 0 Solutions

1 Simulating a Markov chain

We discussed that deterministic initial condition is simply a special case of the stochastic, namely $X_0 = x_0$ deterministically can be modeled using the pmf

$$\mu(x) = \begin{cases} 1 & x = x_0, \\ 0 & \text{else.} \end{cases}$$

1.1 Suppose we have a pmf over $\mathcal{X} = \{x_1, \dots, x_k\}$. Then it can be specified as a k -tuple $(p_1, \dots, p_k)$ where

$$\mu(x) = \begin{cases} p_1 & x = x_1 \\ \dots & \\ p_k & x = x_k \end{cases}$$

where $p_1, \dots, p_k$ are non-negative and sum to 1.

We discussed one way to sample from μ :

1. Split the interval into disjoint regions $R_0, \dots, R_k$ of size $p_0, \dots, p_k$ respectively.^a
2. Sample $U \sim \text{Unif}(0, 1)$, i.e. generate a number uniformly at random between 0 and 1.
3. If U is in R_j , then return x_j .

^a. We can only do this because the “sizes” are non-negative and add up to 1, which is guaranteed by the properties of the pmf.

In passing, we mentioned that this is simply a special case of the inverse sampling method, which we recalled for probability density functions that are strictly positive (so that the cdf was invertible)

1. Given a probability density function p^* over $\mathbb{R}$, compute the cdf

$$F: \mathbb{R} \rightarrow [0, 1]$$
$$F(x) = \int_{-\infty}^x p^*(s) ds$$

and subsequently (or directly without computing the cdf) the inverse of the cdf $F^{-1}(u): (0, 1) \rightarrow \mathbb{R}$.

2. Sample $U \sim \text{Unif}(0, 1)$.
3. Return $F^{-1}(U)$.

However this method only works for one-dimensional densities.^b

^b. Can you think of a way to extend the inverse sampling technique to some density $p^*(x_1, \dots, x_n)$ over $\mathbb{R}^n$?

1.2 (Rejection Sampling) We defined the random variable A as

$$A = \begin{cases} 1 & U \leq \frac{p^*(X)}{Mp_0(X)} \\ 0 & \text{else.} \end{cases}$$

Since X is a continuous random variable, the event $\{X = x\}$ has probability 0 for all $x \in \mathbb{R}$. So we consider the conditional cdf

$$F_{X|A=1}(x) = \mathbb{P}(X \leq x | A = 1) = \frac{\mathbb{P}(X \leq x, A = 1)}{\mathbb{P}(A = 1)}.$$

We then calculated the joint probability

$$\begin{aligned} \mathbb{P}(X \leq x, A = 1) &= \mathbb{P}\left(X \leq x, U \leq \frac{p^*(X)}{Mp_0(X)}\right) \\ &= \int_{-\infty}^x \int_{-\infty}^{p^*(s)/Mp_0(s)} f_{X,U}(s, u) du ds \\ &= \int_{-\infty}^x \int_{-\infty}^{p^*(s)/Mp_0(s)} p_0(s) \mathbb{1}_{(0,1)}(u) du ds \\ &= \int_{-\infty}^x \int_0^{p^*(s)/Mp_0(s)} p_0(s) du ds \\ &= \int_{-\infty}^x p_0(s) \frac{p^*(s)}{Mp_0(s)} ds \\ &= \frac{1}{M} \int_{-\infty}^x p^*(s) ds. \end{aligned}$$

X and U are independent, and continuous random variables. Hence they are jointly continuous with joint density

$$f_{X,U}(x, u) = f_X(x)f_U(u)$$

where $f_X = p_0$ and $f_U = \mathbb{1}_{(0,1)}$.

Here we explicitly use the fact that $p^*(x) \leq Mp_0(x)$ for all $x \in \mathbb{R}^d$, and that the density $\mathbb{1}_{(0,1)}$ is unity in $(0, 1)$ and zero outside.

Now we can easily compute $\mathbb{P}(A = 1)$ by taking $\lim_{x \rightarrow \infty}$ on both sides, i.e. marginalizing over X .

$$\mathbb{P}(A = 1) = \frac{1}{M}$$

This allows us to compute the condition cdf

$$F_{X|A=1}(x) = \int_{-\infty}^x p^*(s) ds$$

and so

$$f_{X|A=1}(x) = p^*(x).$$

We discussed the implications of the acceptance rate being $1/M$, and how we could find an appropriate base density p_0 for densities with bounded support, i.e. choose p_0 to be the uniform distribution over the support of p^* .

1.3 Note that $P(x, \cdot)$ is the pmf of the next state conditioned on the current state being x . So starting with a sample x_0 from μ , we can sample x_{n+1} from $P(x_n, \cdot)$ for $n \geq 0$.

2 Markov chain proofs

2.1 As a simpler case, we considered $m = n + 2$. Since we couldn't find a way to simplify $\mathbb{P}(X_{n+2} = y | X_n = x, H_n = h_n)$ without X_{n+1} , we decided to introduce it using the law of total probability

$$\begin{aligned}
\mathbb{P}(X_{n+2} = y \mid X_n = x, H_n = h_n) &= \sum_{y_1 \in \mathcal{X}} \mathbb{P}(X_{n+2} = y, X_{n+1} = y_1 \mid X_n = x, H_n = h_n) \\
&= \sum_{y_1 \in \mathcal{X}} \mathbb{P}(X_{n+2} = y \mid X_{n+1} = y_1, X_n = x, H_n = h_n) \mathbb{P}(X_{n+1} = y_1 \mid X_n = x, H_n = h_n) \\
&= \sum_{y_1 \in \mathcal{X}} \mathbb{P}(X_{n+2} = y \mid X_{n+1} = y_1) \mathbb{P}(X_{n+1} = y_1 \mid X_n = x) \\
&= \sum_{y_1 \in \mathcal{X}} \mathbb{P}(X_{n+2} = y \mid X_{n+1} = y_1, X_n = x) \mathbb{P}(X_{n+1} = y_1 \mid X_n = x) \\
&= \sum_{y_1 \in \mathcal{X}} \mathbb{P}(X_{n+2} = y, X_{n+1} = y_1 \mid X_n = x) \\
&= \mathbb{P}(X_{n+2} = y \mid X_n = x)
\end{aligned}$$

The extension to general $m > n$ was left as homework.

2.2 (Random walks) We related the Bernoulli and Rademacher distributions as follows

$$\begin{aligned}
\xi &\sim \text{Bernoulli}(p) \\
Z &= 2\xi - 1 \implies Z \sim \text{Rademacher}(p).
\end{aligned}$$

So if $\xi_1, \xi_2, \dots \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$, then we can equivalently write the random walk S_n as

$$S_n = \sum_{i=1}^n (2\xi_i - 1) = 2 \sum_{i=1}^n \xi_i - n$$

We also know that if $B_n = \sum_{i=1}^n \xi_i$ where $\xi_i \sim \text{Bernoulli}(p)$, then $B_n \sim \text{Binomial}(n, p)$:

$$\mathbb{P}(B_n = k) = \binom{n}{k} p^k (1-p)^{n-k},$$

for $k \in \{0, \dots, n\}$.

And so

$$\begin{aligned}
\mathbb{P}(S_n = x) &= \mathbb{P}(2B_n - n = x) = \mathbb{P}\left(B_n = \frac{n+x}{2}\right) \\
&= \begin{cases} \binom{n}{\frac{n+x}{2}} p^{\frac{n+x}{2}} (1-p)^{\frac{n-x}{2}} & n+x \text{ is even, } x \in \{-n, \dots, n\} \\ 0 & \text{else} \end{cases}
\end{aligned}$$

The condition that $n+x$ must be even for positive probabilities occurs because we start ($n=0$) at an even number 0 and at each step add an odd increment ± 1 which makes the possible states switch from even to odd and odd to even.

Notice that the probability of a given trajectory ending in $S_n = x$ is just $p^{\frac{n+x}{2}} (1-p)^{\frac{n-x}{2}}$, and the number of such distinct paths is $\binom{n}{\frac{n+x}{2}}$.

2.3 We noted that for a process X_n to be Markov, it must satisfy the Markov condition

$$\mathbb{P}(X_{n+1} = y \mid X_n = x, H_n = h_n) = \mathbb{P}(X_{n+1} = y \mid X_n = x)$$

for any feasible history (h_n, x) . But to find a counter-example to prove that a process is not Markov, it is usually only possible to consider some small number of past states, i.e., a conditional probability like

$$\mathbb{P}(X_{n+1} = y \mid X_n = x, X_{n-1} = x_{n-1}, X_{n-2} = x_{n-2})$$

and show that this quantity depends x_{n-1} and x_{n-2} .¹

Consider $M_n = \max\{0, S_n\}$. If $M_n = k$, where $k > 0$, then we know that $S_n = k$. On the other hand if $M_n = 0$, then $S_n \leq 0$, i.e.

$$\{M_n = 0\} = \{S_n \leq 0\}, \quad \{M_n = k\} = \{S_n = k\} \quad (k > 0).$$

We discussed that it is best to check for counterexamples where n is small and the state is close to zero, so let us consider

$$\mathbb{P}(M_3 = 1 \mid M_2 = 0, M_1 = 1, M_0 = 0) \stackrel{?}{=} \mathbb{P}(M_3 = 1 \mid M_2 = 0, M_1 = 0, M_0 = 0).²$$

It can be checked that this equality does not hold.

$$\begin{aligned} \mathbb{P}(M_3 = 1 \mid M_2 = 0, M_1 = 1, M_0 = 0) &= \mathbb{P}(S_3 = 1 \mid S_2 \leq 0, S_1 = 1, S_0 = 0) \\ &= \mathbb{P}(S_3 = 1 \mid S_2 = 0, S_1 = 1, S_0 = 0) \\ &= \mathbb{P}(S_3 = 1 \mid S_2 = 0) \\ &= p \end{aligned}$$

$$\begin{aligned} \mathbb{P}(M_3 = 1 \mid M_2 = 0, M_1 = 0, M_0 = 0) &= \mathbb{P}(S_3 = 1 \mid S_2 \leq 0, S_1 \leq 0, S_0 = 0) \\ &= \mathbb{P}(S_3 = 1 \mid S_2 \leq 0, S_1 = -1, S_0 = 0) \\ &= \mathbb{P}(S_3 = 1 \mid S_2 \in \{0, -2\}, S_1 = -1, S_0 = 0) \end{aligned}$$

We can't directly use Markov property here, so we need to try to break up $\{S_2 \in \{0, -2\}\}$ into the disjoint events $\{S_2 = 0\}$ and $\{S_2 = -2\}$

$$\begin{aligned} &= \frac{\mathbb{P}(S_3 = 1, S_2 \in \{0, -2\} \mid S_1 = -1, S_0 = 0)}{\mathbb{P}(S_2 \in \{0, -2\} \mid S_1 = -1, S_0 = 0)} \\ &= \frac{\mathbb{P}(S_3 = 1, S_2 = 0 \mid S_1 = -1, S_0 = 0) + \mathbb{P}(S_3 = 1, S_2 = -2 \mid S_1 = -1, S_0 = 0)}{\mathbb{P}(S_2 = 0 \mid S_1 = -1, S_0 = 0) + \mathbb{P}(S_2 = -2 \mid S_1 = -1, S_0 = 0)} \\ &= \frac{\mathbb{P}(S_3 = 1, S_2 = 0 \mid S_1 = -1) + \mathbb{P}(S_3 = 1, S_2 = -2 \mid S_1 = -1)}{\mathbb{P}(S_2 = 0 \mid S_1 = -1) + \mathbb{P}(S_2 = -2 \mid S_1 = -1)} \\ &= \frac{\mathbb{P}(S_3 = 1 \mid S_2 = 0, S_1 = -1)\mathbb{P}(S_2 = 0 \mid S_1 = -1) + \mathbb{P}(S_3 = 1 \mid S_2 = -2, S_1 = -1)\mathbb{P}(S_2 = -2 \mid S_1 = -1)}{p + (1 - p)} \\ &= \mathbb{P}(S_3 = 1 \mid S_2 = 0)\mathbb{P}(S_2 = 0 \mid S_1 = -1) + \mathbb{P}(S_3 = 1 \mid S_2 = -2)\mathbb{P}(S_2 = -2 \mid S_1 = -1) \\ &= p \cdot p + 0 \cdot (1 - p) \\ &= p^2 \end{aligned}$$

1. There is no clear process by which one can look at a process and intuit whether it is Markov or not. We have chosen such unclear processes as examples here to demonstrate how you might proceed.

Broadly there is no more general advice we can give than to simplify conditional probabilities in hopes of finding a proof or a counter-example.

2. We noted that checking for $n = 2$ was not useful since the $M_0 = 0$ with probability 1.

Consider only feasible histories
Markov property of S_n
Probability of an increment is p .

So M_n is not a Markov chain in general for any p . We mentioned in class that for $p = 1$ or $p = 0$, the example yields equality for this case. We then noted that these cases correspond to the deterministic sequences $M_n = n$ and $M_n = 0$ respectively, which are Markov trivially.

Next we consider $A_n = |S_n|$. Unlike earlier, each event $\{A_n = x\}$ splits into two sets $\{S_n = x\}$ and $\{S_n = -x\}$.³ So it seems unreasonable to try and write conditional probabilities in terms of S_n as earlier, so we look for a different simplification.

3. We discussed that this is not true for $x = 0$, so we need to consider it as a different case.

We would like to prove

$$\mathbb{P}(A_{n+1} = y \mid A_n = x, H_n = h_n) = \mathbb{P}(A_{n+1} = y \mid A_n = x).$$

Note that for $x = 0$, $\{A_n = 0\} = \{S_n = 0\}$, and so we can simply use the Markov property and the fact that if X and Y are independent random variables, then $f(X)$ and $g(Y)$ are also independent random variables for any measurable function f and g . So we consider $x \neq 0$

Since we can use the Markov property when we condition on S_n , we introduce it using marginalization

$$\begin{aligned} \mathbb{P}(A_{n+1} = y \mid A_n = x, H_n = h_n) &= \sum_{s \in \mathbb{Z}} \mathbb{P}(A_{n+1} = y, S_n = s \mid |S_n| = x, H_n = h_n) \\ &= \mathbb{P}(A_{n+1} = y, S_n \in \{x, -x\} \mid |S_n| = x, H_n = h_n) \\ &= \mathbb{P}(A_{n+1} = y, S_n = x \mid |S_n| = x, H_n = h_n) + \dots \end{aligned}$$

where $\dots$ is the same expression but with the event $\{S_n = -x\}$.

We consider the term

$$\begin{aligned} \mathbb{P}(A_{n+1} = y, S_n = x \mid |S_n| = x, H_n = h_n) &= \mathbb{P}(A_{n+1} = y \mid S_n = x, |S_n| = x, H_n = h_n) \mathbb{P}(S_n = x \mid |S_n| = x, H_n = h_n) \\ &= \mathbb{P}(A_{n+1} = y \mid S_n = x, H_n = h_n) \mathbb{P}(S_n = x \mid |S_n| = x, H_n = h_n) \\ &= \mathbb{P}(A_{n+1} = y \mid S_n = x) \mathbb{P}(S_n = x \mid |S_n| = x, H_n = h_n) \end{aligned}$$

(and similarly for the other term with $\{S_n = -x\}$)

Here the first term does not depend on h_n . So if we can remove the dependence of the second term on h_n , then we are done. We can further simplify

$$\begin{aligned} \mathbb{P}(S_n = x \mid |S_n| = x, H_n = h_n) &= \frac{\mathbb{P}(S_n = x, |S_n| = x, H_n = h_n)}{\mathbb{P}(|S_n| = x, H_n = h_n)} \\ &= \frac{\mathbb{P}(S_n = x, H_n = h_n)}{\mathbb{P}(S_n \in \{-x, x\}, H_n = h_n)} \\ &= \frac{\mathbb{P}(S_n = x, H_n = h_n)}{\mathbb{P}(S_n = x, H_n = h_n) + \mathbb{P}(S_n = -x, H_n = h_n)} \end{aligned}$$

We recall that the marginal probability $\mathbb{P}(S_n = x)$ is $N_x p_x$ where $N_x = \binom{n}{\frac{n+x}{2}}$ is the number of trajectories ending at x in the n -th step, and the probability of each trajectory is $p_x = p^{\frac{n+x}{2}} (1-p)^{\frac{n-x}{2}}$.

We can then compute $\mathbb{P}(S_n = x, H_n = h_n)$, we multiply p_x by the number of trajectories ending at x in the n -th step where the absolute values of the trajectory $(|x_0|, \dots, |x_{n-1}|)$ is h_n .⁴ Call this number $N_x^{h_n}$.

4. Can you justify why this should be true?

Similarly we can compute $\mathbb{P}(S_n = -x, H_n = h_n)$, we multiply p_{-x} by the number of trajectories ending at x in the n -th step where the absolute values of the trajectory $(|x_0|, \dots, |x_{n-1}|)$ is h_n , and we call this number $N_{-x}^{h_n}$.

In doing so, we can write

$$\mathbb{P}(S_n = x \mid |S_n| = x, H_n = h_n) = \frac{N_x^{h_n} p_x}{N_x^{h_n} p_x + N_{-x}^{h_n} p_{-x}}.$$

The final step we discussed was to note that for every trajectory that ended at x , there was a corresponding mirrored trajectory ending at $-x$ both of which satisfy $(|x_0|, \dots, |x_{n-1}|) = h_n$, and vice-versa. So $N_x^{h_n} = N_{-x}^{h_n}$, and the only terms depending on h_n cancel out, and we can write

$$\begin{aligned}\mathbb{P}(S_n = x \mid |S_n| = x, H_n = h_n) &= \frac{p_x}{p_x + p_{-x}} \\ \mathbb{P}(S_n = -x \mid |S_n| = x, H_n = h_n) &= \frac{p_{-x}}{p_x + p_{-x}}.\end{aligned}$$

We are writing this out for completeness. In your exams, if you prove that $\mathbb{P}(A_{n+1} = y \mid A_n = x, H_n = h_n)$ does not depend on the history h_n , then you can simply conclude that A_n is Markov.

that is

$$\begin{aligned}\mathbb{P}(A_{n+1} = y \mid A_n = y, H_n = h_n) &= \mathbb{P}(A_{n+1} = y \mid S_n = x) \mathbb{P}(S_n = x \mid |S_n| = x, H_n = h_n) + \mathbb{P}(A_{n+1} = y \mid S_n = -x) \mathbb{P}(S_n = -x \mid |S_n| = x, H_n = h_n) \\ &= \mathbb{P}(A_{n+1} = y \mid S_n = x) \mathbb{P}(S_n = x \mid |S_n| = x) + \mathbb{P}(A_{n+1} = y \mid S_n = -x) \mathbb{P}(S_n = -x \mid |S_n| = x) \\ &= \mathbb{P}(A_{n+1} = y \mid S_n = x, |S_n| = x) \mathbb{P}(S_n = x \mid |S_n| = x) + \mathbb{P}(A_{n+1} = y \mid S_n = -x, |S_n| = x) \mathbb{P}(S_n = -x \mid |S_n| = x) \\ &= \mathbb{P}(A_{n+1} = y, S_n = x \mid |S_n| = x) + \mathbb{P}(A_{n+1} = y, S_n = -x \mid |S_n| = x) \\ &= \mathbb{P}(A_{n+1} = y \mid A_n = x).\end{aligned}$$

3 Markov Decision Processes

Ask us during TA1 :)